



core EMG

Hardware User Manual

Table of Contents

1	General Warnings and Cautions	5
1.1	Risks and Benefits	5
1.2	Safety Information Summary	5
2	Introduction	7
2.1	Brief Description:	7
2.2	Device Safety Categories	7
2.3	Intended Use	8
2.4	Contraindications	8
3	Definitions	9
3.1	Graphic Symbols and Meaning	9
3.2	Glossary of Terms	10
4	Product Information	11
4.1	Model Designation	11
4.2	Product Versions and Configurations	11
4.3	Software Requirements	11
5	Setting up the Hardware	11
5.1	System Unboxing	11
5.2	Core Hub Overview	12
5.3	Core EMG Sensor Overview	13
5.4	Hardware Setup Instructions	14
5.5	Installing the Companion Software – myoRESEARCH™	14
5.6	Configuring the Hardware	15
6	Basic Operation Instructions	17
6.1	Safety Information Summary	17
6.2	Operating Environment	17
6.3	Ensure the Core EMG Sensor is in Measurement Mode	18
6.4	Attach the Core EMG Sensor to a Patient or Subject	19
6.5	Configure a Measurement	19
6.6	Enter the Measurement Screen	21
6.7	Check for Normal Signal Display with Interface to a PC	21
6.8	Check Signal Quality	21
6.9	Record Signal as Desired	23
6.10	Shutdown After Use	23
6.11	Storage and Protecting Between Use	23
7	Additional Core EMG Settings and Features	23
7.1	Core Hub Synchronization	23

7.2	Lossless Data Recovery	24
7.3	Signal Processing for Amplitude Normalization	24
8	Maintenance	25
8.1	Consumable Items	25
8.2	Maintenance by Users	25
8.3	Companion Software Updates	26
8.4	Device Software (Firmware) Updates	26
8.5	Battery Replacement	27
9	Troubleshooting, Fault Diagnosis	27
9.1	Troubleshooting Chart	27
9.2	Website Link to FAQ	28
9.3	Radio Considerations	28
9.4	Setting the RF Network	28
10	Support, Service, and Repair	29
10.1	Submitting Technical Support Requests	29
10.2	Returning Equipment	29
11	Taking Product out of Operation	29
11.1	Disposal of Equipment and Batteries	29
12	Technical Information	30
12.1	Theory of Operation	30
12.2	Electro-Magnetic Compatibility Tables	31
13	Appendices	34
13.1	Appendix A – Product Specifications	34
13.2	Appendix B – Use of Disposable Electrodes	36
13.3	Appendix C – Radiation Exposure Information Regarding Use of EMG Sensors	37
13.4	Appendix D – Radio Regulatory Statements	38
13.5	Appendix E – Core EMG LED Status Guide	39

- Manufacturer:

Noraxon U.S.A. Inc.
15770 North Greenway-Hayden Loop, Suite 100
Scottsdale, AZ 85260
Tel: (480) 443-3413
Fax: (480) 443-4327
Email: info@noraxon.com
Support Email: support@noraxon.com
Web Site: www.noraxon.com

- Authorized European Representative:

EC	REP	Advena Limited, Tower Business Centre, 2nd Flr., Tower Street, Swatar, BKR 4013 Malta
-----------	------------	--

Website: <http://www.advenamedical.com>

CE Marking and Notified Body:

Clearance to market this product in the European Union and United Kingdom has been certified by Notified Body #2797, British Standards Institution (BSI).

CE 2797

No part of this document may be copied, photographed, reproduced, translated, or reduced to any electronic medium or machine-readable form without prior written consent of Noraxon U.S.A. Inc.

Noraxon and myoRESEARCH are registered trademarks and the Noraxon logo, myoANALOG, myoFORCE, myoMETRICS, myoMOTION, myoMUSCLE, myoPRESSURE, myoVIDEO, myoSYNC, NiNOX, TRUsync and Ultium are common-law trademarks of Noraxon U.S.A., Inc. All other trademarks are the property of their respective owners. ©2018, all rights reserved.

General Warnings and Cautions

1.1 Risks and Benefits

There is **no identified risk of physical harm or injury** with use of the Core EMG System. The benefit provided by use of the device is the provision of objective measures to assess the severity of pathological human movement conditions and gauge any subsequent improvement offered by therapy, training, or design changes.

1.2 Safety Information Summary



Cautions

- Never use the Core EMG equipment on a person with an implanted pacemaker.
- Never operate the Core EMG equipment within one meter of any critical medical device.
- Keep small cells and batteries which are considered swallowable out of the reach of children.
 - Swallowing may lead to burns, perforation of soft tissue, and/or death. Severe burns can occur within two hours of ingestion.
 - Seek medical attention immediately in case of ingestion of a cell or battery.



Warnings

- Do not immerse the Core EMG equipment in any water or liquid.
- Do not drop the Core EMG equipment.
- Do not use in conditions where the temperature can exceed 55 °C (130 °F).
- Do not use the Core EMG equipment on individuals undergoing MRI, Electro Surgery or Defibrillation.
- The Core EMG product produces results that are informative, not diagnostic. Qualified individuals must interpret the results.
- Do not modify Core EMG equipment without authorization of the manufacturer.



Attention

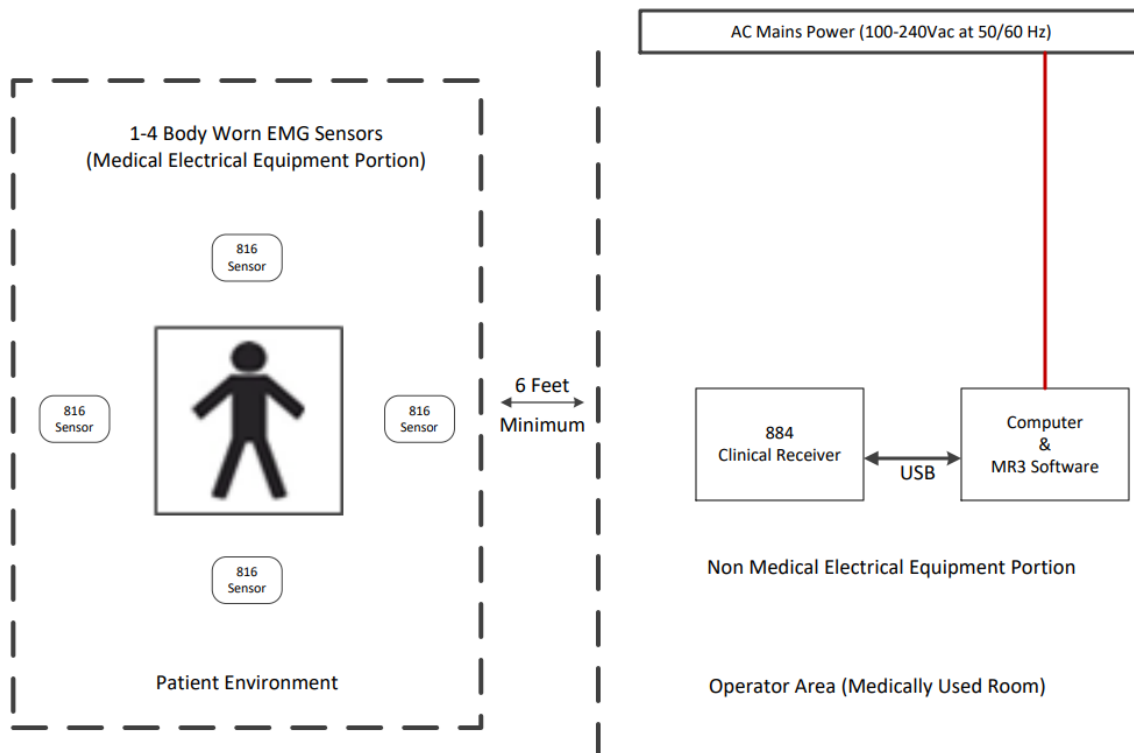
- The operator must be familiar with typical characteristics of the signals acquired by the Core EMG equipment and be able to detect anomalies that could interfere with proper interpretation.
- Any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority of the Member State in which the user and/or patient is established.
- Instructions for Use are to be provided in electronic form unless specifically requested. Should the user like to obtain a hard copy of these Instructions, please contact the manufacturer.
- Power supply should be positioned in a manner that ensures accessibility.



Guidance on Conditions for Safe Use

The Core EMG sensors are suitable for use as medical devices. To ensure safe operation the following guidelines must be strictly followed.

- The Core Hub and computer must not be accessible (touchable) by the patient. Ensure the Core Hub and computer are always positioned at least six feet (two meters) away from the patient.
- The Core Hub must only be connected to a certified ITE/medical grade computer or devices (If ITE certified computer is used, then basic safety at a system level shall be verified).
- To ensure safe operation, the use of Core EMG with custom built computers (those assembled from discrete subassemblies) must be avoided.
- The operator must never touch the patient and any other device (Core Hub or computer) at the same time.



Core Ultium Medical Electrical System Diagram

2 Introduction

2.1 Brief Description:

The Core EMG System wirelessly transmits surface electromyography data from the electrode/sensor site to a desktop transceiver.

With the wireless sensor attached directly to the electrode, collecting EMG measurements is greatly simplified by eliminating any cable connections and the need for additional disposables, i.e., sensor tape. This concept gives the user the flexibility to operate the Core EMG System through a streamlined data collection process and without the traditional laboratory-based measurement limitations. The small, lightweight EMG sensors are also beneficial for small subjects like children and small animals.

The Core EMG System is designed to operate any configuration between 1 and 4 sensors through the accompanying Windows-based software (MR-myomUSCLE) or mobile application (Ultium EMG Biofeedback for Android).

The default system is equipped with EMG sensors, but a synchronization capability can be used to accurately synchronize the Core EMG System to other biomechanical measurement systems integrated into the MR software, e.g., video, IMUs, force plates, and pressure distribution platforms.

2.2 Device Safety Categories

The Core EMG System incorporates the following ratings:

CISPR 11 Class A radio frequency compatibility (ref. Section 12 for details)

Note: The EMISSIONS characteristics of this equipment make it suitable for use in industrial areas and hospitals (CISPR 11 class A). If it is used in a residential environment (for which CISPR 11 class B is normally required) this equipment might not offer adequate protection to radio-frequency communication services. The user might need to take mitigation measures, such as relocating or re-orienting the equipment.

2.3 Intended Use

Noraxon's Core EMG sensor is intended to measure and quantify muscle biopotential signals separately or in combination with other kinematic or kinetic signals. These inter-related measurements can provide data to assist in understanding movement abnormalities. The system is designed as a measuring tool for clinical, investigational, and research applications. Additionally, this device is compliant to the international standard for diagnostic EMG (IEC 60601-2-40).

Intended Users

Researchers or individuals trained in physical medicine, physical therapy, human performance or ergonomics and who have gained familiarity with the Core EMG System and its software.

Subject Populations – Medical

Individuals with neurological disorders, physical injuries, pre/post-surgical or post stroke conditions.

Subject Populations – Non-medical

Athletes, workers at their work site, subjects in new product trials.

Common Applications

Gait analysis; identification of inconsistencies and abnormalities; tracking over time the outcome of surgical, therapeutic, or orthotic interventions; identification of ergonomic stress factors in the workplace or new product designs.

Suitable Environments

The Core EMG System is suitable for use in hospitals, clinics, academic laboratories except near active HF SURGICAL EQUIPMENT or MRI IMAGING EQUIPMENT.

Expected (Essential) Performance

The Core EMG System (EMG sensor and Hub) synchronously acquires and displays muscle activity signals from multiple muscle sites in real time.

Consideration should be given to commonplace wireless networks that may be present in the environment. Under the conditions of excessive wireless communications performance of the Core EMG System may exhibit sluggish (non-real time) behavior.

Risk-Benefit

There is **no identified risk of physical harm or injury** with the use of the Core EMG product. The benefit provided by use of the device is the provision of objective measures to assess the severity of pathological human movement conditions and gauge any subsequent improvement offered by therapy, training, or design changes.

Special Concerns

The Core EMG System operates by means of microwave radio frequency transmissions. Certain (older, vintage model) pacemaker devices may be susceptible to such microwave transmissions. Therefore, use of the device is contra-indicated in individuals who have implanted pacemakers.

2.4 Contraindications

Use of the Core EMG System is contra-indicated in individuals who have implanted pacemakers.

3 Definitions

3.1 Graphic Symbols and Meaning

The following international icons and symbols are found on the Core EMG enclosures and in this user manual. Their meaning is described below.



Approval to market this product (#816) in the European Community was certified by Notified Body #2797 BSI.



The device generates radio frequency energy during operation.



A 5 Volt DC external power source is applied to this connection.



The device is suitable for a direct electrical attachment to the body.



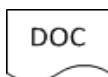
Read material in the Instruction Manual wherever this symbol appears.



Identifies the manufacturer of the device.



Identifies the serial number of the device.



Additional information available in a separate document

3.2 Glossary of Terms

Core EMG Sensor -- A small individual radio transmitter worn on the body used to measure and transmit electromyography (EMG) signals. The Core EMG Sensor is directly applied to the skin by an adhesive dual electrode.

Core Charging & Receiving Hub -- A USB-connected device that charges and receives wireless signals from one to four Core EMG sensors.

Core EMG Serial Number -- A unique five-character tag used to identify each EMG Sensor. The members of any Core EMG network are determined by their serial numbers.

Probe -- A generic term for any EMG Sensor.

RF -- (Abbreviation for Radio Frequency) Wireless communication takes place on assigned radio frequencies or channels. For the Core EMG System, RF transmissions occur at frequencies between 2.4 GHz and 2.5 GHz. Other wireless systems including WiFi and Bluetooth commonly operate at the same frequencies and can be a source of interference.

BLE -- (Abbreviation for Bluetooth Low Energy) The power-conserving variant of Bluetooth personal area network technology that uses frequency hopping wireless technology in the 2.4 GHz unlicensed radio band to interconnect nearby devices.

RF Network -- RF transmissions for the Core EMG System can be selected to occur on one of 4 different networks. All networks use frequency hopping to avoid interference.

RF Traffic -- The presence of radioactivity present on a given frequency like the number of cars on an expressway. Several users (wireless devices) may be communicating using the same frequencies. The best operation of the Core EMG System occurs when the RF Traffic is low.

4 Product Information

4.1 Model Designation

The basic Core EMG System consists of two primary components:

- Model 884 Core Hub (1 per system)
- Model 816 Core EMG Sensor (up to 4 per system)

4.2 Product Versions and Configurations

The model 884 Core Hub provides charging and can receive wireless transmission for up to four Core EMG Sensors.

4.3 Software Requirements

The Core EMG System requires software to perform its function. The equipment is offered in combination with the following software programs:

- Noraxon myoRESEARCH® (MR3.20 or newer) Software Platform and myoMUSCLE™ Software Module
- Ultium EMG Biofeedback App for Android



5 Setting up the Hardware

5.1 System Unboxing

The Core EMG System is packed within a reinforced padded box for storage and protection during transport. Upon receiving the package, carefully remove all contents and verify the following components are present.



Figure 1 – Core Hub (#884)



Figure 2 – Core EMG Sensor (#816)



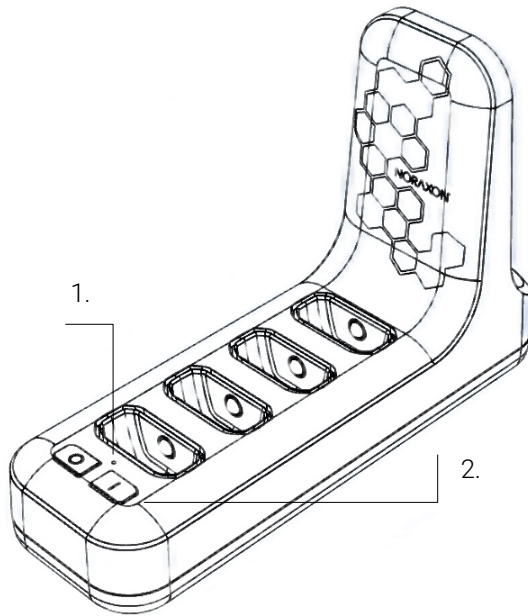
Figure 3 – 5V USB-C Power Supply & Cable (#PSU11 & #CBL44)

Note: The number of EMG sensors will vary.

Additional items that may be included with your Core EMG System include:

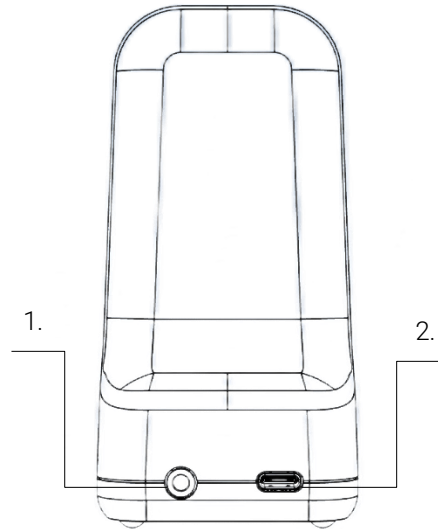
- Dual EMG electrodes (#271S)
- Core EMG User Manual (#P-8848)

5.2 Core Hub Overview



Hub (Front/Top)

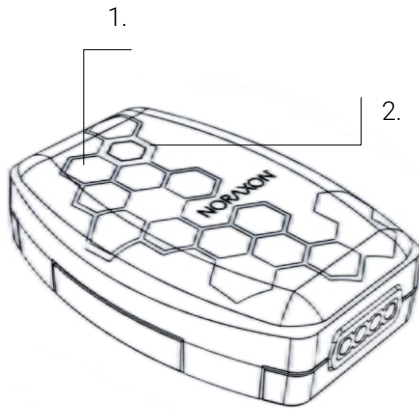
1. **Status LED** – Indicates the status of the hub.
2. **OFF/ON Power Buttons** – Sends Power OFF or ON command to all sensors inserted into the hub.



Hub (Back)

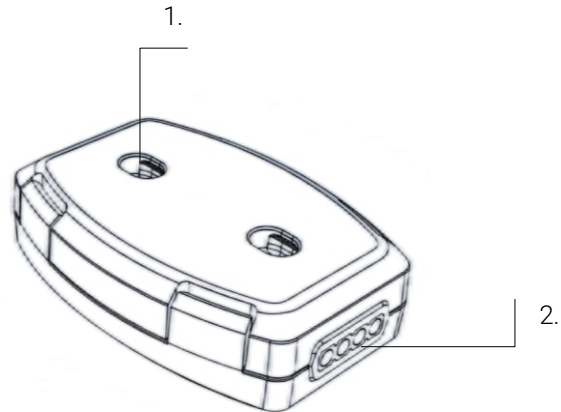
1. **Sync Port** – TTL (on-off) compatible 3.5 mm audio jack connection to other devices.
2. **USB-C Port** – USB connection between the computer and hub.

5.3 Core EMG Sensor Overview



EMG Sensor (Front/Bottom)

1. **Power Button** – Power the sensor On/Off. Hold for 3+ seconds for a hard reset.
2. **Status LED** – The LED indicator flashes BLUE when awaiting connection, flashes GREEN when measuring through wireless, and flashes LIGHT BLUE when measuring through Bluetooth LE. The LED indicator shows solid AMBER when charging.



EMG Sensor (Back/Bottom)

1. **Electrode Snaps** – Third electrode that provides a ground reference for the differential signals collected by the EMG leads attached to the surface electrode.
2. **Charger Contacts** – Sensor battery is charged, and sensor data is exchanged through these points.

5.4 Hardware Setup Instructions

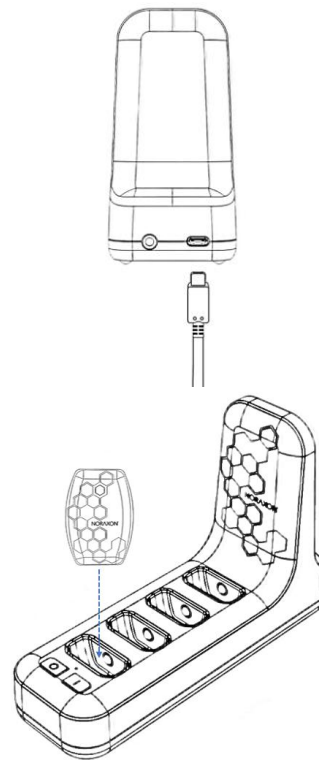
Step 1

Insert the USB-C cable (#CBL44) into the USB connector on the back of the Core Hub (#884).

Insert the opposite end of the USB-C cable into an available USB-C port on the computer.

Step 2

Insert the Core EMG Sensors (#816) into the Hub.



5.5 Installing the Companion Software – myoRESEARCH™

To utilize the full functionality of the Core EMG system, and ensure the system has updated drivers, Noraxon's *myoRESEARCH* (MR) needs to be installed on the computer.

Note: The Core Hub requires the Noraxon USB device driver which is pre-installed by the MR software installation. It is also available in the [Downloads](#) section of the Noraxon website.

Software Installation

Within the Core EMG System shipment, there is a USB flash drive containing the latest *myoRESEARCH* software.

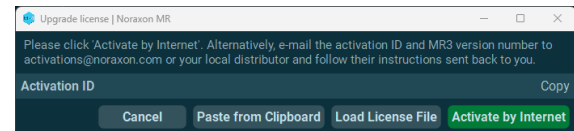
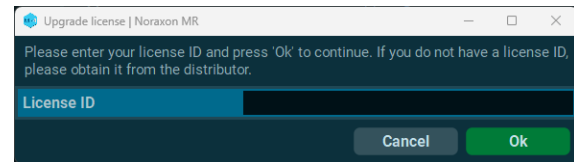
1. Insert the MR USB flash drive into the PC
2. Open the installer application (e.g., noraxon.mr.4.0.0) and follow the instructions
3. After installation, an icon will be created on the desktop



Companion Software Activation

The companion software must be *activated* before unrestricted use is possible.

1. Open myoRESEARCH
2. A dialog box will indicate how many more times MR can be opened in trial mode
3. Click **Activate**
4. Enter the License ID provided on your USB flash drive and click **Ok**
5. If you have an internet connection, click **Activate by Internet** for immediate activation
 - a. Alternatively, email the provided activation ID to activation@noraxon.com
 - b. Noraxon Support will email or respond by phone with the Activation Code
 - c. Enter the provided Activation Code to remove any restrictions on use

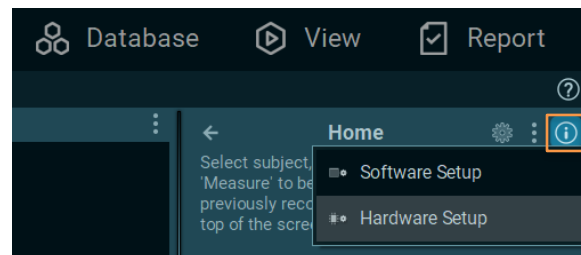


5.6 Configuring the Hardware

Before the Core EMG system can be used, the device software settings must be configured to recognize the different components that make up the system. Follow the instructions below to update the hub firmware, sensor firmware, and populate sensors to prepare for data collection.

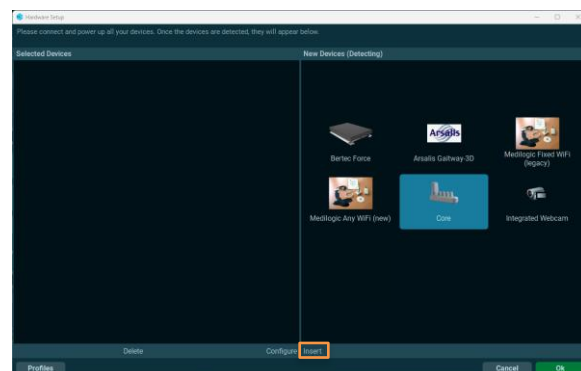
Step 1

Open the MR software, click on the settings icon (⚙️) in the upper right-hand corner, then click **Hardware Setup**.



Step 2

After connecting the Core Hub to a USB port on the computer, select the Core Hub icon in the **New Devices** area, and click **Insert**.



Step 3

The Core EMG hardware setup window will appear as shown.

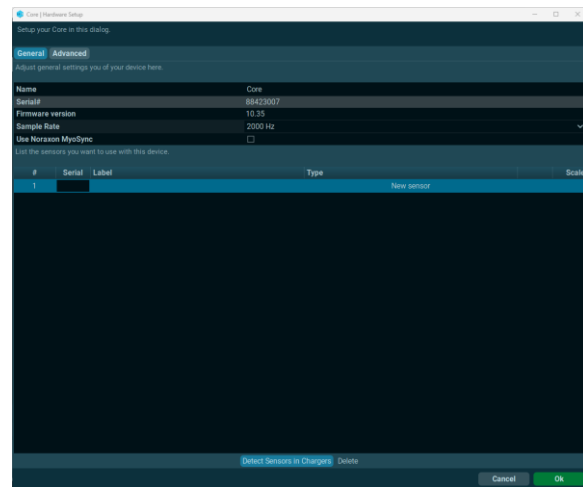
Within the “General” tab, select the desired settings for the system, i.e., name, sample rate.

Note: If available, firmware updates for the Receiver and Sensors will be indicated here.

When using the myoSYNC, click the **Use Noraxon myoSYNC** checkbox.

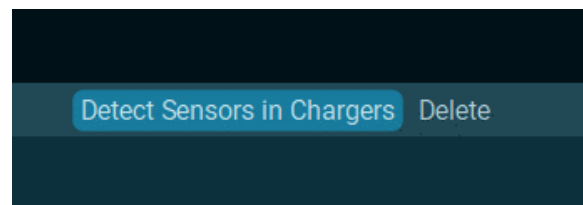
Additional settings, i.e., RF network, are available within the “Advanced” tab.

Note: If multiple Core or Ultium systems are in use, separate RF networks assignments are required for each Receiver.

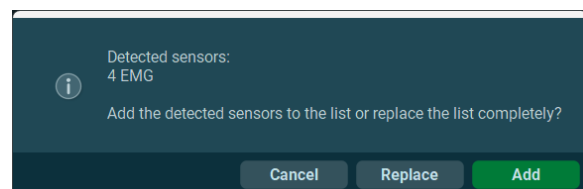
**Step 4**

Make sure the Core EMG Sensors are placed in the Hub, then click **Detect Sensors in Chargers** to load all sensor serial numbers into the MR software.

Note: The first serial number in the list is associated with the Sensor closest to the front of the Charger (i.e., toward the Charger LEDs).

**Step 5**

After detection is complete, click **Replace** to replace all existing sensor serial numbers associated with this device or **Add** to append additional sensor serial numbers to an already existing list.



List the sensors you want to use with this device.

#	Serial	Label	Type
1	40013	EMG 1	EMG
2	40014	EMG 2	EMG
3	40019	EMG 3	EMG
4	40029	EMG 4	EMG

Step 6

Click **Ok** to save the hardware settings and close the setup window.

Cancel

Ok

6 Basic Operation Instructions

6.1 Safety Information Summary

Strictly follow all safety practices given in Section 1 of this manual. The most critical safety practices are repeated here.

**CAUTION**

- Never use the Core EMG System on a person with an implanted pacemaker
- Never operate the Core EMG System within 1 meter of any critical medical device

6.2 Operating Environment

The software is structured into 3 levels or tiers:

1. Microsoft Windows Operating System
2. myoRESEARCH (MR) – A common multi-device service module running under Windows for management of various specific types of devices and their related software modules
3. myoMUSCLE – One of the software modules running under MR that is specific to EMG devices such as the Core EMG System

The Core EMG System (a member of the Ultium product family) acquires EMG signals from the patient via EMG sensors (the medical device) which wirelessly transmit the measured signals to any receiver that is part of the Ultium product family. In the case of the Core Hub, the receiver is attached to a general-purpose computer via a USB connection which may include an optional (non-medical) USB hub. The computer is a certified ITE/medical grade computer operating Windows 10 or higher. The accompanying MR-myoMUSCLE software combination records and displays the EMG signals during a measurement session and provides analysis and reporting.

Required Computer Specifications

The following are the current specifications for computer systems needed to run the various software programs that are distributed by Noraxon. These specifications are subject to change without notice due to the volatility of the computer industry. It may be assumed that systems shipped by Noraxon will meet or exceed these specifications.

Note: Systems quoted with these specifications are valid for 90 days from the date of issue.

O/S: Windows 11, 64-bit

Processor: Intel® Core™ i7 Processor or equivalent (8th Generation, PassMark score of 10000 or better)

Memory: 16 GB RAM minimum

Hard Drive: 256 GB SSD (M2 NVMe, large capacity (1 TB) optional if there is a secondary hard drive)

Graphics: A GPU with a PassMark score of 3000 or better is required. Nvidia Quadro P600 or GTX 1050 GPU or better is recommended.

Note: A discrete graphics processor (Nvidia, Radeon, AMD, etc.) is recommended to run MR. An integrated graphics processor (i.e. Intel Integrated Graphics) may not be powerful enough.

OpenGL: Version 4.6 or higher

Open CL: Required

Ports:

- 1 or more USB Type-C
- 2 or more USB Type-A

Note: Noraxon supplied USB cables must be used.

Camera: Logitech C920 (or equivalent DirectShow compatible webcam) for use as a reference camera.

These computer specifications must be met for all customers who are purchasing myoRESEARCH and are required for multi-device systems due to the increased demand on the computer system. It is also recommended that an external hard drive be available for data backup.

Lastly, although your computer may have the hardware indicated above, it is highly recommended that you update your graphic board drivers using the manufacturer's website.

6.3 Ensure the Core EMG Sensor is in Measurement Mode

Check the Sensor Status LED

The sensor's LED status indicator provides a means of communicating its operational state.

In the idle state, the status indicator will flash BLUE. When the sensor is actively measuring over the wireless hopping network, the status indicator will flash GREEN. When the sensor is actively measuring over a Bluetooth LE connection, the status indicator will flash LIGHT BLUE.

When a sensor is charging, the status indicator will display solid AMBER.

If the status indicator is not illuminating at all, the EMG Sensor may be powered off or may have a depleted battery.

6.4 Attach the Core EMG Sensor to a Patient or Subject



The sensor is secured to the skin by the disposable adhesive electrode. It is recommended to reinforce sensor attachment during highly dynamic activities by wrapping material around the sensor (e.g., Coban) or adding tape over the top (e.g., Cover-Roll Stretch).

Attach the Core EMG sensor to a dual adhesive electrode, then attach the sensor-electrode combination to the desired muscle site.

Note: The sensor should be attached to the electrode on a hard, flat surface to ensure proper connection before being attached to the measurement site.



Proper operation requires the EMG Sensor to be applied to the measurement site so that the adhesive side of the dual electrode is in direct contact with bare skin. Some skin preparation around the measurement site may be beneficial if the EMG signal exhibits a wandering baseline. (See Appendix A)

Note: The Core EMG Sensor is designed for use with the dual EMG electrodes provided by Noraxon with the standardized 20 mm inter-electrode spacing.

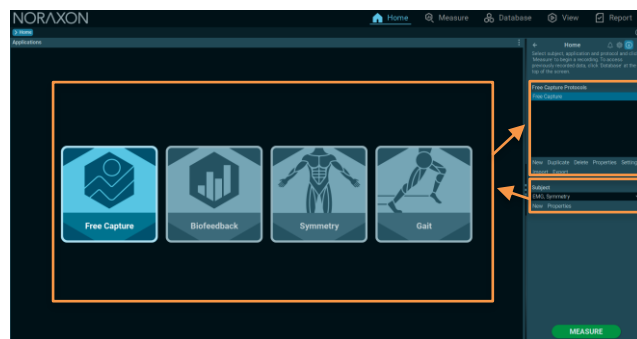
6.5 Configure a Measurement

Step 1

On the **Home** screen, select or create a subject.

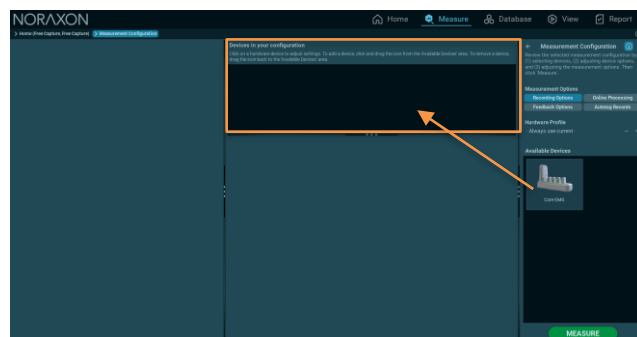
Select an application, then select or create a new protocol.

Click **MEASURE** to proceed.



Step 2

Insert the Core EMG device into the configuration by dragging it from the **Available Devices** list to the **Devices in Your Configuration** area.

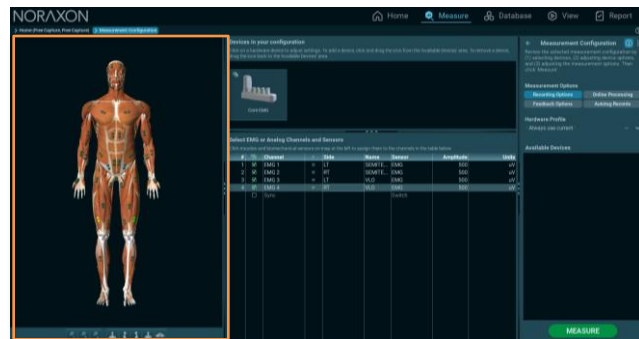


Step 3

Select the channels to include in the measurement by clicking the checkmark.

While a channel is selected, assign a label by selecting the corresponding muscle on the **Muscle Map** or entering the name manually.

Click **MEASURE** to proceed.

**Step 4**

Confirm signal quality and click **RECORD** to begin your recording data.

**Step 5**

Click **STOP** to end the recording, then enter a name for the new record.

Click **Save & View** to view and begin processing the record, or one of the other options (**Discard & Measure Again** or **Save & Measure Again**) to return to the measurement screen.

Please enter name for the new record.

Name:

Subject:

New

Discard & Measure Again Save & Measure Again Save & View

6.6 Enter the Measurement Screen

Continue to the next step by selecting **Measure**.

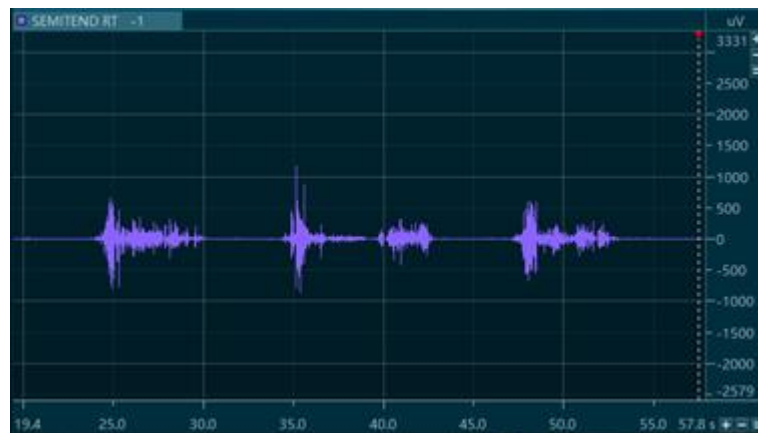
Exceptional Functions/Situations (error messages)

Error Message	Could not start all sensors
Meaning	One or more EMG sensors failed to respond to a start measurement command. Check to see if any selected EMG sensors are powered off. This can also happen if the subject is more than 30 m away from the Core Hub.

Note: The Hub operates utilizing power provided through the USB connection to a computer. Certain power saving modes employed by modern computers may disable USB ports when a period of inactivity is detected. If a USB shutdown occurs the Hub power indicator will be extinguished, and the system will fail to respond. It is recommended that the power settings on the computer be checked and if needed changed to allow continuous operation of all USB ports.

6.7 Check for Normal Signal Display with Interface to a PC

When used with the companion software, the Core EMG System displays and records raw (or processed) EMG waveforms, like the waveform displayed below.



Consult the myoRESEARCH Software User Manual for additional instructions and descriptions for recording, playback, and processing of the data acquired by the Core EMG system.

6.8 Check Signal Quality

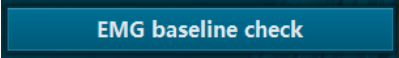
Electromyography recordings depend on many best practice standards to ensure ideal signal quality. Within the Noraxon myoRESEARCH platform, standard procedures and best practice functionality has been built in to assist the practitioner with identifying their current signal quality. With the Core EMG system, a live signal status monitor provides real-time information for practitioners to understand the current recording conditions.

EMG Signal Status Monitor

The Status Monitor appears at the top right-hand corner of the measurement screen to show relevant EMG baselines before recording. RMS voltage will appear in μV to ensure the signal is at a low resting amplitude before recording. Median frequency will appear in Hz to indicate any electromagnetic field interference from mains AC power. Impedance will appear in $\text{k}\Omega$ to check for high levels of impedance that may interfere with signal quality.

EMG Signal Status Monitor On/Off

Select **EMG Baseline Check** to show or hide the EMG Status Monitor within the measurement screen.



Red/Green Indicator

The circular red/green indicator will show if there are issues with the signal that will interfere with the quality of the recording. You can hover over it with the mouse for a full explanation of possible issues.

Green Indicator

The indicator will remain **GREEN** when the impedance, RMS voltage, and frequency is sufficient to produce a clean signal.



Red Indicator

The indicator will turn **RED** if:

1. The impedance is $> 100 \text{ k}\Omega$
2. The RMS value is $> 5 \text{ }\mu\text{V}$ and the median frequency of the signal is 50 Hz or 60 Hz ($\pm 3 \text{ Hz}$). This is because these are the frequencies of mains AC power which can interfere with the EMG signal.



Related Software Measurement Tools

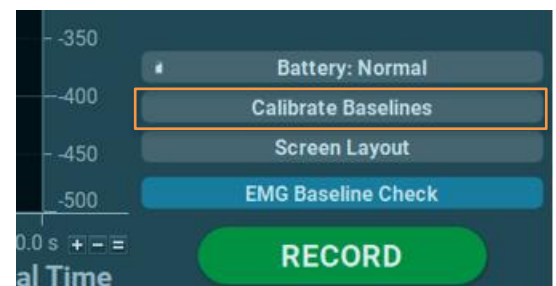
Within the Noraxon myoRESEARCH software is a suite of tools that can be used during EMG measurements. Below is a list of the most used features for electromyographic recordings. Additional information around more advanced features can be found on www.noraxon.com

Baseline Calibration

Data collected during the first second of a measurement is used to correct for any offset present in the signal. During the first second of each measurement, ensure that the subject is not performing highly dynamic movements to capture a clean signal recording.

To perform this step manually, select **Calibrate Baselines**.

Choose **EMG Baseline Check** to show the baseline RMS voltage in the Status Monitor.



CAUTION

This tool will adjust baselines for all EMG and analog channels simultaneously.

6.9 Record Signal as Desired

After checking the quality of each signal, click **Record** at the top left of the screen to begin recording a measurement and click **Stop** to end the recording. Label the record as the name of the configuration or type a new name into the field, then save the recording by choosing **Save & Measure**, **Save & View**, or **Discard & Measure Again**.

6.10 Shutdown After Use

At the end of use:

- Insert all Core EMG sensors into the slots on the Core Hub
- Push the Sensor Power button on the front of the Hub to power all sensors off.

6.11 Storage and Protecting Between Use

For extended storage or when travelling:

- Insert all sensors into the Hub and power them off.
- Position all components inside the protective packaging according to the prepared cavities.
- Every 3 months, fully charge the sensors once to continue storage without connection to power.

Note: It is necessary to charge the sensors once every 3 months. The Lithium Ion batteries will expand if left discharged for too long.

7 Additional Core EMG Settings and Features

7.1 Core Hub Synchronization

To synchronize the signal(s) from the Core EMG System with other devices, it is necessary to configure the hardware and software to accept an incoming sync signal.

Sync In Port

The Sync In port on the back of the Hub is designed to accept a 2-5 V TTL sync input via 3.5 mm audio jack.

MR Sync Setup

Sync for the Core Hub is enabled by clicking the *Use Noraxon MyoSync* checkbox in the hardware setup menu.

Core | Hardware Setup

Setup your Core in this dialog.

General Advanced

Adjust general settings you of your device here.

Name	Core
Serial#	88423015
Firmware version	10.52
Sample Rate	2000 Hz
Use Noraxon MyoSync	☒

When *Use Noraxon MyoSYNC* is enabled, the digital channel will be labeled Sync. When it is not enabled, the channel will be labeled Switch and can be used as a generic digital input.

Select EMG or Analog Channels and Sensors								
Click muscles and biomechanical sensors on map at the left to assign them to the channels in the table below.								
#	☒	Channel	※	Side	Name	Sensor	Amplitude	Units
1	☒	EMG 1	※			EMG	500	uV
2	☒	EMG 2	※			EMG	500	uV
3	☒	EMG 3	※			EMG	500	uV
4	☒	EMG 4	※			EMG	500	uV
5	☒	Sync				Switch	2	On

7.2 Lossless Data Recovery

The Core EMG System supports “lossless” data recovery. The sensors continue to capture and store data (>8 hours of data) when communication is temporarily lost with the Hub. Once the record is completed, data can be recovered in real-time or later using the import function.

Real-Time (Online) Recovery

Immediately following the record, MR will begin wirelessly recovering any data that may have been lost. It is recommended to instruct the subject to stay within, or return to, a 2–5-meter radius of the Hub during the recovery process. If online recovery is terminated, data can be recovered later via offline data recovery.

Offline Data Recovery (Import Function)

Data can be recovered at a later point in time via the import function. When ready to recover data (offline) for one or multiple records, go to the Database tab within MR, then click ‘Import’ on the right-side tool menu. The following options will be displayed:

- Clicking ‘Offline Data Recovery’ will display a list of records that have incomplete data.
- Select each record from the list to recover data.
- Ensure that all sensors are inserted into the slots on the Hub and the Hub is connected to the PC.
- Click OK to initiate the data recovery process.

Import from External Location
 Import Video Files
 Import MRXP Data
 Import Record from a Text File
 Import Record from a C3D File
 Import Record from Noraxon DataLogger (*.tm2)
 Import Zebris16 Data
 Import MyoPressure Data
 Import Record from MyoMotion DataLogger (*.mml)
 Import Record from Ultium DataLogger (*.nudl)
Offline Data Recovery

Note: Offline data recovery will occur approximately 20 times faster than online recovery.

7.3 Signal Processing for Amplitude Normalization

The companion software program, MR, supports real-time and post-processing amplitude normalization of the EMG signal. To view the instructions and options for completing this process in MR, refer to the supplemental MR training material at www.noraxon.com.

8 Maintenance

8.1 Consumable Items

The consumable items listed here are designed for one-time use. It is suggested to regularly restock to have available product when needed.

Part No.	Image	Description
271S/271SCASE		Dual solid gel EMG electrodes (30 per pouch or 210 per box)

8.2 Maintenance by Users

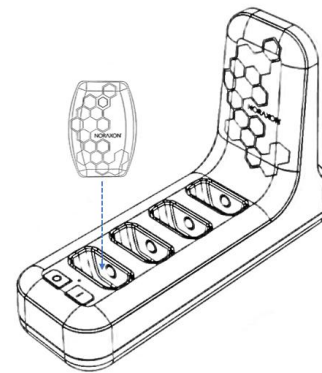
Routine maintenance is recommended for the Core EMG system for successful performance. Cleaning and charging the EMG sensors regularly will help preserve long-term operation.

Charging the EMG Sensors

Note: No safety precautions are required for charging the EMG sensors.

Step 1

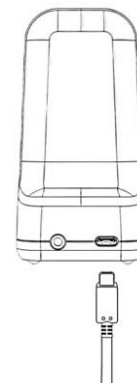
Verify that all the sensors (#816) are correctly inserted into the Core Hub (#884).



Step 2

Insert USB-C cable into the port on the back of the Core Hub (#884). Connect the other end of the USB-C cable to the PC or a compatible USB power adapter.

- Verify that the LED indicator on all sensors shows solid AMBER.
- Charge for approximately 3 hours or until the LED indicator turns off.



Cleaning The EMG Sensors

For sanitary purposes, it is advisable to clean the back of the Core EMG sensors on a regular basis. Core EMG sensors can be cleaned with a cloth slightly dampened with a solution of mild soap and water. Use of even mild disinfectant solutions (i.e., isopropyl alcohol swabs or household disinfectant wipes) is not recommended as they can degrade polypropylene plastics in the sensor housing over time.

The EMG Sensors are not warranted against exposure to any of the conventional forms of sterilization (e.g., autoclave, heating, etc.). Users wishing to utilize this equipment in a sterile environment, such as an operating theater, should consult Noraxon for other options.



CAUTION

- Only use a damp cloth with mild soap and water to clean the bottom of the EMG Sensors.
- **DO NOT** immerse EMG Sensors in any water or liquid.

Storing the EMG Sensors

It is recommended to store the sensors in the Hub while connected to power to maintain charge during storage. The sensors can be disconnected from power for a maximum of 3 months.

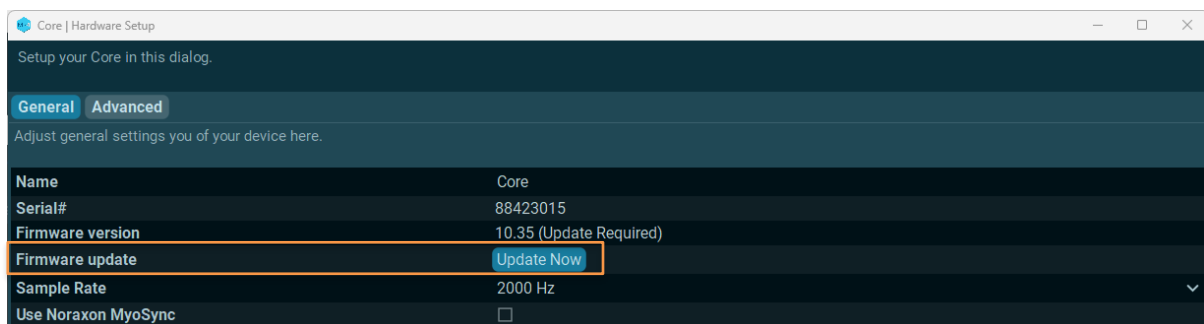
Note: It is necessary to charge the sensors once every 3 months to avoid battery expansion.

8.3 Companion Software Updates

- Perform a backup of the data folders to a separate drive as a precaution.
- Click on the Patch/Update link provided in the email or as given on the [Noraxon website](#).
- Download the Patch/Update file.
- To install the Patch/Update, click "Run" on the dialog box and follow the instructions provided within the installation wizard.

8.4 Device Software (Firmware) Updates

The internal program (firmware) for the Core EMG devices can be updated via MR. The user will be notified in the hardware setup if an update is required.



CAUTION

All EMG sensors should be fully charged before a firmware update is performed.

8.5 Battery Replacement

Battery Life Expectations

The Lithium-Ion coin cell battery used in the Core EMG sensors is rated for a minimum of 300 charge-discharge cycles. Typical usage is 500 charge-discharge cycles. As the number of charge-discharge cycles increases the battery capacity slowly declines thereby reducing run time despite being fully charged.

Brand new batteries can operate for up to 4 hours when fully charged. If the run time of the sensors drops to <2 hours, battery replacement should be considered.

Battery Replacement Procedures

The coin cell batteries may not be replaced by the user. Only qualified Noraxon technical personnel may perform maintenance.

Initiate battery replacement by completing a submission form at www.noraxon.com/battery-replacement.

9 Troubleshooting, Fault Diagnosis

9.1 Troubleshooting Chart

Symptom: Problem with the PC recognizing the Core EMG System

Possible Reason

USB cable is disconnected or loose

USB Driver is not installed

Remedial Action

Check USB cable connection at both Hub and computer

Check the Device Manager for USB driver errors and update the driver if needed

Symptom: Problems with EMG Sensors communicating with the Core Hub

Possible Reason

Sensors were not assigned to Hub

RF traffic is blocking data transmission

1+ sensor(s) are not turned on

Remedial Action

Assign sensors (see Section 5.6)

Change the RF channel in hardware setup

Make sure sensor is charged and on

Symptom: Problems with individual EMG Sensors

Possible Reason

Sensor was not assigned to Hub

Sensor battery is low (or sensor does not flash)

Sensor shifts with very dynamic movements

Remedial Action

Assign sensor (see Section 5.6)

Retry after charging sensor for at least 15 minutes

Secure sensor with overlying elastic wrap

*Symptom: Problems with intermittent EMG Sensor signals***Possible Reason**

Electrode is loose or disconnected

Electrode gel is displaced or expired

Sensor is too far from Hub

Sensor radio signal is partially blocked (absorbed) by subject's body (esp. at long distances)

Remedial Action

Check electrode connection at the sensor and muscle site

Use a new electrode

Move to within 30 m of Hub

Reposition sensor on subject to obtain a direct line-of-sight relationship between sensor and Hub

9.2 Website Link to FAQ

Answers to frequent questions can be found at Noraxon's Frequently Asked Questions (FAQ) website page at this link:

<https://www.noraxon.com/support-learn/technical-support/faqs/>

Other educational material is available at this link:

<https://www.noraxon.com/support-learn/technical-support/>

9.3 Radio Considerations

The Core EMG radio system operates in the 2400 MHz ISM (Industrial, Scientific, and Medical) radio band reserved for use in most countries of the world. The radio transfers data digitally using a proprietary wireless sensor protocol. Other devices operating in this frequency band include computer networks, microwave ovens, cordless phone sets and other WiFi enabled devices.

Despite all this competing radio activity the Core EMG System can discern its information from all the surrounding radio traffic. Reliable transmission depends on good signal quality. Signal quality will fall with extended distances between the Hub and the EMG Sensors. Obstructions (walls, metal structures, trees, etc.) between the Hub and the EMG Sensors will also lower the signal quality.

The EMG sensors contain FLASH memory which backs up all measured data during a recording. The companion software can query the FLASH memory to recover data that was lost to interference.

While the Core EMG system is quite immune to interference, it does transmit a deliberate radio signal that could affect nearby sensitive equipment. Users should always be aware of this possibility. In a similar manner, although the energy level of the radio is considered harmless to human beings, it is still prudent to minimize exposure.

Finally, although available worldwide, each country places certain restrictions on the operation of radios in the 2400 MHz ISM band. These restrictions include allowable transmitter power levels and broadcast frequencies

9.4 Setting the RF Network

The Core EMG radio system utilizes a proprietary frequency hopping protocol for sensor communication. Three RF Network addresses may be optionally selected to allow co-location of multiple Ultium or Core systems. All 3 networks hop on the same frequencies, but the hopping order is unique to each network, allowing them to co-exist without interfering with each other.

10 Support, Service, and Repair

10.1 Submitting Technical Support Requests

A Support Request can be submitted using the online form available at this link:

<https://www.noraxon.com/support-learn/support-request/>

Provide all information requested by the form including a **detailed** description of the problem being experienced and your telephone number or e-mail address.

10.2 Returning Equipment

Be sure to obtain an RMA Number (return material authorization) before returning any equipment. Completing the online service request form will assign an RMA Number. Otherwise contact Noraxon USA.

<https://www.noraxon.com/support-learn/rma-request/>

Send the equipment **postage prepaid** and **insured** to the address below. Include the RMA Number on the shipment label. Mark the package "Goods to be repaired – Made in USA" to avoid unnecessary customs charges. (Beware listing a Customs or Insurance value of \$5,000.00 USD or more will result in a delay at United States Customs.)

Noraxon USA
15770 N. Greenway-Hayden Loop
Suite 100
Scottsdale, AZ
85260, USA

If you are shipping from outside the USA please use UPS, FedEx, DHL, or EMS (US Postal Service) and **not a freight-forwarder**. Using a freight-forwarder incurs additional brokerage fees. If a package is shipped to Noraxon via a carrier other than the ones listed above, it may be refused.

11 Taking Product out of Operation

11.1 Disposal of Equipment and Batteries

The Core EMG Sensors contain Li-Ion coin cell batteries, which may be hazardous if disposed of incorrectly. Please check with the governing authorities in your location before disposing of the Core EMG System and its contents.

12 Technical Information

12.1 Theory of Operation

The Core EMG wireless system is based on a pre-certified transceiver module. This radio module operates in the 2.4 GHz bands with an output power level of < 100 mW and is based on a proprietary frequency hopping protocol.

Model 816 Core EMG Sensor

Each Core EMG transmitter module (#816) incorporates one transceiver module together with an analog front end and supporting circuitry. The #816 EMG Sensor is powered by one 150mAh battery; each is identified by a unique serial number.

The #816 EMG Sensor contacts the patient only by the disposable dual EKG style electrode. Two recessed snap receptacles on the back of the #816 EMG Sensor enclosure are designed to attach to the dual EKG style electrode, which then secures the sensor-electrode system to the patient. The dual electrode is intended to be in contact with bare skin.

The bottom end of the #816 EMG Sensor has 4 contact pads for recharging its battery and communication. To recharge the battery, the #816 EMG Sensor is placed inside a fitted slot on the charging and receiving hub. The device cannot be applied to the patient/subject and charged at the same time.

Model 884 Core Hub

The Core Hub (part #884) is configured to hold up to four #816 EMG Sensors. All battery-charging controls are inside the sensor modules. The sensor charger merely supplies a 5VDC source of power through a set of spring-loaded pins. The spring-loaded pins contact the charging pads of the #816 EMG Sensor. The 5VDC supply is from a USB connection to a PC or a medical grade external power supply by Globtek (model GTM46161-155.0-USBC).

The Core Hub (part #884) also consists of a main motherboard and a radio transceiver module. The Core Hub has **no applied parts**.

The Core Hub interfaces to a PC and is powered via a USB-C port.

The transceiver in the #884 Core Hub can communicate with up to four (4) #816 EMG Sensors.

12.2 Electro-Magnetic Compatibility Tables

Guidance and manufacturer's declaration – electromagnetic emissions		
The Core EMG System is intended for use in electromagnetic environment specified below. The customer or the user of the Core EMG System should assure that it is used in such an environment.		
Emissions Test	Compliance	Electromagnetic environment - guidance
RF emissions CISPR 11	Group 2	The Core EMG System must emit electromagnetic energy in order to perform its intended function. Nearby electronic equipment may be affected.
RF emissions CISPR 11	Class A	The Core EMG System is suitable for use in all establishments other than domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic Emissions IEC 61000-3-2	Not applicable	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Not applicable	

Guidance and manufacturer's declaration – electromagnetic immunity			
The Core EMG System is intended for use in electromagnetic environment specified below. The customer or the user of the Core EMG System should assure that it is used in such an environment.			
Immunity Test	IEC 60601 test level	Compliance level	Electromagnetic environment - guidance
Electrostatic discharge (ESD) IEC 64000-4-2	±8 kV contact ± 15 kV air	±8 kV contact ±15 kV air	Device user should avoid touching subject and sensor probes while a measurement is active.
Electrical fast transient/burst IEC 61000-4-4	±2kV for power supply lines ±1kV for input/output lines	±2kV for power supply lines Not applicable	For battery charging mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1kV differential mode ±2kV common mode	±1kV differential mode ±2kV common mode	For battery charging mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5 % U_T (>95 % dip in U_T) for 0,5 cycles 40 % U_T (60 % dip in U_T) for 5 cycles 70 % U_T (30 % dip in U_T) For 25 cycles <5 % U_T (>95 % dip in U_T) For 5 secs	Not applicable to operation Not applicable to operation Not applicable to operation Not applicable to operation	For battery charging mains power quality should be that of a typical commercial or hospital environment.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE U_T is the a.c. mains voltage prior to application of the test level.			

Guidance and manufacturer's declaration – electromagnetic immunity			
The Core EMG System is intended for use in electromagnetic environment specified below. The customer or the user of the Core EMG System should assure that it is used in such an environment.			
Immunity Test	IEC 60601 test level	Compliance level	Electromagnetic environment - guidance
Conducted RF IEC 61000-4-6 (Charging System) Radiated RF IEC 61000-4-3	3 Vrms 150 kHz to 80 MHz 3 V/m 80 MHz to 2,5 GHz	3Vrms 3V/m	Portable and mobile RF communications equipment should be used no closer to any part of the Core EMG System, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d = 1.2\sqrt{P}$ $d = 1.2\sqrt{P} \quad 80 \text{ MHz to } 800 \text{ MHz}$ $d = 2.3\sqrt{P} \quad 800 \text{ MHz to } 2,5 \text{ GHz}$ where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,^a should be less than the compliance level in each frequency range.^b Interference may occur in the vicinity of equipment marked with the following symbol: 
NOTE 1 At 80 MHz and 800 MHz, the higher frequency range applies.			
NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			
^a Field strengths from fixed transmitters, such as a base station for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the Core EMG System is used exceeds the applicable RF compliance level above, the Core EMG System should be observed to verify normal operation. If abnormal operation is observed, additional measures may be necessary, such as reorienting or relocating the Core EMG System. ^b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.			

Recommended separation distances between portable and mobile RF communications equipment and the Core EMG System			
The Core EMG System is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Core EMG System can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the Core EMG System as recommended below, according to the maximum output power of the communications equipment.			
Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter m		
	150 kHz to 80 MHz	80 MHz to 800 MHz	800 MHz to 2,5 GHz
	$d = 1.2\sqrt{P}$	$d = 1.2\sqrt{P}$	$d = 2.3\sqrt{P}$
0,01	0.12	0.12	0.23
0,1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
NOTE 1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			
NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

13 Appendices

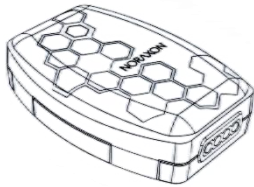
13.1 Appendix A – Product Specifications

Expected Useful Lifetime

Both the Core Hub (#884) and Core EMG Sensor (#816) have a usable life of seven years.

The EMG sensors (#816) operate with a rechargeable Lithium-Ion coin cell battery. The battery capacity will decline with ongoing use and require replacement after 300+ discharge/charge cycles to preserve the device's rated 4 hours of operating time.

Dimensions and Weight



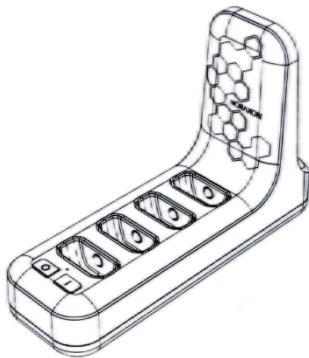
Core EMG Sensor

Dimensions

1.65" L x 1.18" W x 0.49" H
(42 mm L x 30 mm W x 12.5 mm H)

Weight:

0.53 oz (15 g)



Core Hub

Dimensions

6.69" L x 2.36" W x 4.45" H
(170 mm L x 60 mm W x 113 mm H)

Weight

5.11 oz (145 g)

Performance Characteristics

Output & Transmission Frequency (Depending on country)

- Up to 100 mW (depending on country allowance)
- GFSK 2402-2480 MHz, frequency hopping
- Utilizing up to 79 hopping channels
- Channel-to-channel hopping frequency 50 Hz (20 msec)
- EMG sensor transmission range: 30 m (typical)

EMG Sensor Data Acquisition System

- 16-bit resolution
- Sample Rate: 2000 Hz

EMG Specifications

- Snap-style terminal electrode connections
- No notch (50/60 Hz) filters are used
- High-pass cutoff: 10 Hz
- Low-pass cutoff: 500 Hz
- Internal 24-bit sampling
- Baseline noise < 2 μ V RMS
- Input impedance > 100 MOhm
- CMR > 90 dB
- Input range: +/- 24,000 μ V
- Adaptive Resolution:
 - < +/- 5,000uV: 0.3 μ V
 - > +/- 5,000uV: 1.16 μ V
- Measurement Functional Accuracy: +/-1% F.S.
- Sensor operation up to 4 hours on a fully charged battery (recharge time 3 hours)
- 1 Gigabit embedded memory (up to 8 hours of internal storage)

Energy Consumption, Condition of Use

- Core Hub is powered by USB connection to PC

Environmental Conditions for Normal Operation

- Ambient Temperature: 0 °C to 38 °C
- Relative Humidity: 95% RH
- Atmospheric Pressure: 70 kPa to 107 kPa

Environmental Conditions for Storage and Transport

- Ambient Temperature: -40 °C to +70 °C
- Relative Humidity: <95% RH
- Atmospheric Pressure: 70 kPa to 107 kPa

13.2 Appendix B – Use of Disposable Electrodes

While the Core EMG sensors can operate with reusable electrodes, it is typically used with disposable surface electrodes. Any good quality silver/silver chloride dual electrode with 20 mm inter-electrode spacing is acceptable. Noraxon provides quality disposable electrodes for a wide variety of Surface EMG applications. Other electrodes may be used, but it is recommended that any electrodes used with the Core EMG sensors satisfy the requirements for standard ANSI/AAMI EC12-1991 Disposable ECG electrodes.



- Because disposable electrodes have a shelf life, it is important not to use expired parts.
- Bulk disposable electrodes come packaged in a sealed container or bag.
- The expiration date can be found printed on the package container.
- After the sealed bulk container is opened, the remaining electrodes should be used before their gel begins to dry out.
- Always keep the remaining electrodes in their bulk package until they are used.
- If the electrode package does not seal itself, closing the package with tape or using a zippered plastic bag is recommended.
- Do not store the electrode package in the direct sun, as this will accelerate drying.
- Avoid using electrodes that are randomly found lying outside of their bulk packaging as their expiration date is uncertain and their gel has been exposed to accelerated drying.

Be aware that when disposable electrodes are removed, some individuals may notice a faint red skin discoloration over the site previously occupied by the electrode. This skin discoloration is typically benign and temporary and may be due to a mild allergic reaction to the adhesive or simply be a slight abrasion caused by peeling away the tape. It will usually disappear within 24 hours.

Noraxon discourages any attempt to reuse a disposable electrode, even if it is simply pulled off to slightly reposition the electrode's muscle placement. Some of the electrode gel may remain on the original site and the EMG signal may be affected. Also, sometimes the electrode adhesive may not adhere to the skin as well when it is reapplied. Noraxon strongly recommends against the use of dried out electrodes that are re-wetted with electrode gel.

Electrode Application Guidelines and Facts

- If the subject has a fair amount of hair at the electrode application site, the hair should be clipped. Shaving is not necessary and may irritate the skin.
- The electrode application site should be clean and dry. The preferred method of cleaning is with soap and water plus drying the skin with a dry cloth. Dry skin contributes to good electrode adhesion and good trace quality.
- Cleaning with isopropyl alcohol should be limited to situations where electrode adhesion is an issue (diaphoresis, excessively oily or lotion covered skin), since it may dehydrate the skin thereby causing skin impedance to increase. If alcohol is used, allow it to dry prior to electrode application.
- Noraxon recommends attaching the lead wire to the electrode prior to placing the electrode on the skin. This will eliminate the potential for discomfort if snap lead wires are pressed onto the electrode after the electrode has been applied. It will also prevent the electrode gel from seeping out. Additionally, this method will prevent unattached leads from coming into accidental contact with other conductive objects.
- Electrode application sites may need to be abraded to lower the skin impedance. Fine sandpaper or electrode prep gel, e.g., NuPrep, can be used to abrade the skin.

- Electrodes are the weak link in the EMG measurement chain. Lack of proper attention to electrode quality or site preparation is by far the most common cause of inferior recordings.
- It may take up to 5 minutes for disposable electrodes to fully stabilize electrically once applied to the skin. If extremely critical or precise measurements are intended, the electrodes should be applied several minutes in advance of the recording.

13.3 Appendix C – Radiation Exposure Information Regarding Use of EMG Sensors

Each EMG sensor contains a radio frequency transmitter. The radiated power emitted from each individual EMG sensor is very low. To put this in perspective, at full power each EMG sensor transmits at less than 0.8% of the power of a typical active cell phone. Radiation exposure from a single EMG sensor is thus extremely low.

The EMG sensors are designed to operate at two different power levels to keep the already very low levels of radiation exposure to an absolute minimum. The EMG sensors activate their higher power level only during periods of actual data collection. During idle times (at setup and in between actual measurements) the EMG sensors reduce their radiated power to an even lower level (less than 0.01% of the power of a typical active cell phone).

The effects of non-ionizing radiation on biological tissue are still being studied and published 'safe levels' of exposure are subject to review. Today, cell phone usage is widespread and declared 'safe,' although the long-term cumulative effect of cell phone usage has yet to be determined. In contrast, the EMG sensors operate at power levels 125 to 10,000 lower than typical cell phones while limiting exposure to a single episode over a brief time interval.

Because there can be multiple EMG sensors applied in intimate contact with the body, their total collective radiation effect may be questioned. Based on comparative power levels, a full complement of 4 EMG sensors emits a combined (distributed) radiation level still an order of magnitude lower than that of a typical cell phone, which radiates all its energy from one focal point (next to the person's head).

At present, Noraxon identifies no restrictions on use and placement of the EMG sensors on any portion of the human body. The EMG sensors operate at radio frequencies known to affect older style pacemakers. Because the effects are not known at this time, Noraxon advises against using the EMG system on anyone with an implanted pacemaker.

In summary it is prudent to keep in mind that due to biological diversity, certain individuals may have higher sensitivity to radiated emissions. Although it has never been known to occur, the use of the EMG system should be stopped if the person being monitored reports any unusual sensations.

13.4 Appendix D – Radio Regulatory Statements

FCC Statement

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation. Caution: Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device contains modules with FCC ID: IWC-800.

Industry Canada Statement

This product contains a radio module with Canadian Cert No IC: 9113A-800

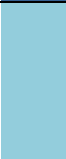
13.5 Appendix E – Core EMG LED Status Guide

The Core EMG sensor and Core Hub have an LED indicator which displays the status of the sensor and hub. Please reference the following list for information on what each LED status means.

Sensor:

Color	Status Description
	Continuous flashing dark blue Sensor is on but not measuring data
	Continuous flashing green Sensor is on and actively measuring data in MR
	Continuous flashing light blue Sensor is on and actively measuring data via Bluetooth OR attempting to recover lost packets during <i>Online Data Recovery</i>
	Single red flash Flashes when the Sensor is turned off
	Single green flash Flashes when the Sensor is turned on
	Solid amber Indicates charging. If the Amber LED turns off while the sensor is still plugged into the charging station, this means that the sensor is fully charged.
	Flash white 8 times Sensor is found by "Find My Sensor" feature in the measurement configuration and can collect data
	Flash green then immediately to red and powers off When turning on, this means that the sensor charge level is too low.
	Continuously flashing green and amber Sensor firmware cannot be found. Please contact Noraxon Support (support@noraxon.com)

Hub:

Color	Status Description
	Continuous flashing dark blue Hub is powered/on
	Flashing light blue A firmware update may not have been completed. Remove the USB cable and plug it in again. Wait 60 seconds for the light to change to dark blue. If the light does not change to flashing dark blue after 3 repeated tries, please contact Noraxon Support (support@noraxon.com).
	Solid red Another Core Hub or Ultium Receiver may be using the same RF Network

NORAXON®